Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Solution of $(x+1)\frac{dy}{dx}+1=e^{x-y}$ is

(A) $e^y(x+1)=c$

(B) $e^y(x+1)=e^x+c$

(C) $e^y(x+1)=ce^x$

(D) none of these

Q.2

The differential equation representing the family of curves $y^2=2c(x+\sqrt{c})$, where c is a positive parameter, is of

(A) order 1 and degree 3

(B) order 2 and degree 4

(C) order 2 and degree 3

(D) order 1 and degree 4

Q.3

The differential equation of all non-horizontal lines in a plane is

(A) $\frac{d^2y}{dx^2}=0$

(B) $\frac{d^2x}{dy^2}=0$

(C) $\frac{dy}{dx}=0$

(D) $\frac{dx}{dy}=0$

Q.4

Solution of the differential equation $x\cos x\left(\frac{dy}{dx}\right)+y(x\sin x+\cos x)=1$ is

(A) $xy=\sin x+c\cos x$

(B) $xy\sec x=\tan^2 x+c$

(C) $xy+\sin x+c\cos x=0$

(D) none of these

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Q.5

The order and degree of the differential equation of all tangent lines to the parabola $x^2 = 4y$ is

(A) 1,2

(B) 2,2

(C) 3,1

(D) 4,1

Q.6

Solution of the equation $(\mathrm{x} + 2\mathrm{y}\hat{}3)\,\mathrm{dy}/\mathrm{dx} - \mathrm{y} = 0 (y > 0)$

(A) $x = y^3 + Ay$

(B) $y(1 - xy) = Ax$

(C) $x(1 - xy) = Ay$

(D) $x(1 + xy) = Ay$

Q.7

$x = 1 + \left(xy\frac{dy}{dx}\right) + \frac{x^2y^2}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{(xy)^3}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots\ldots$ is

(A) $y = \ln x + c$

(B) $y = (\ln x)^2 + c$

(C) $y = \pm\sqrt{(\ln x)^2 + c}$

(D) none of these

Q.8

The differential equation of all parabolas whose axes are parallel to y-axis is

(A) $\frac{d^3y}{dx^3} = 0$

(B) $\frac{d^2x}{dy^2} = C$

(C) $\frac{d^3y}{dx^3} + \frac{d^2x}{dy^2} = 0$

(D) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = C$

Q.9

The degree of the differential equation $\frac{\mathrm{d}^3y}{\mathrm{d}x^2} + 5\left(\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right)^2 = x^3 \log \frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is

(A) 1

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(B) 2

(C) 3

(D) none of these

Q.10

A particular solution of $\log \frac{\mathrm{d}y}{\mathrm{d}x} = 4x + 5y, \quad y(0) = 0$ is given by

(A) $5e^{4x} + 4e^{-5y} = 9$

(B) $5e^{4x} + 4e^{5y} = 9$

(C) $4e^{4x} + 5e^{-4x} = 20$

(D) $4e^{4x} + 5e^{-4y} = 9$

Q.11

If $y + \frac{d}{dx}(xy) = x(\sin x + \log x)$, then

(A) $y = \cos x + \frac{2}{x}\sin x + \frac{2}{x^2}\cos x + \frac{x}{3}\log x - \frac{x}{9} + \frac{c}{x^2}$

(B) $y = -\cos x - \frac{2}{x}\sin x + \frac{2}{x^2}\cos x + \frac{x}{3}\log x - \frac{x}{9} + \frac{c}{x^2}$

(C) $y = -\cos x + \frac{2}{x}\sin x + \frac{2}{x^2}\cos x - \frac{x}{3}\log x - \frac{x}{9} + \frac{c}{x^2}$

(D) None of these

Q.12

The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with

(A) variable radii and a fixed centre $(0, 1)$

(B) variable radii and a fixed centre $(0, -1)$

(C) fixed radius 1 and a variable centres along the x-axis

(D) fixed radius 1 and variable centres along the y-axis

Q.13

The population $p(t)$ at time t of a certain mouse species follows the differential equation $\frac{dp(t)}{dt} = 0.5$ $\mathrm{p}(\mathrm{t}) - 450$. If $\mathrm{p}(0) = 850$, then the time at which the population becomes zero is

(A) $\log 9$

(B) $\frac{1}{2}\log 18$

(C) $\log 18$

(D) $2\log 18$

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Q.14

The solution of the equation $\frac{dy}{dx} = \cos(x-y)$ is

(A) $y + \cot\left(\frac{x-y}{2}\right) = c$

(B) $x + \cot\left(\frac{x-y}{2}\right) = c$

(C) $x + \tan\left(\frac{x-y}{2}\right) = c$

(D) none of these

Q.15

The solution of the equation $\frac{xy'-y}{x} = \tan\frac{y}{x}$ is given by

(A) $\sin\frac{y}{x} = Cx$

(B) $\cos\frac{y}{x} = Cx$

(C) $\tan\frac{y}{x} = Cx^2$

(D) $\tan\frac{y}{x} = Cx^2 + x$

Q.16

The solution of initial value problem $xy' + y = 0, y(2) = -2$ is given by

(A) $xy + 4 = 0$

(B) $(y+2) = 0$

(C) $x - 2 = y + 2$

(D) $(x-2)(y+2)^2 = 4$

Q.17

An equation of the curve in which subnormal varies as the square of the ordinate is (k is constant of proportionality)

(A) $y = Ae^{kx}$

(B) $y = e^{kx}$

(C) $y^2/2 + kx = A$

(D) $y^2 + kx^2 = A$

Q.18

The solution of $y' + \cos(x+y) = \cos(x-y)$ and $y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$ is given by

(A) $\sin y = 2\sin x\cos^2 y$

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(B) $-\cos y = 2\sin x \sin^2 y$

(C) $-\cos y = 2\cos x \sin^2 y$

(D) $\sin x = 2\cos y \sin^2 y$

Q.19

The trajectories orthogonal to $(2a - x)y^2 = x^3$ is

(A) $x^2 + y^2 = C\left(y^2 + 2x^2\right)$

(B) $\left(x^2 + y^2\right)^2 = Cxy$

(C) $\left(x^2 + y^2\right)^2 = C\left(y^2 + 2x^2\right)$

(D) $\left(x^2 + y^2\right) = x^3 + xy^2$

Q.20

General solution of $\frac{dy}{dx} = \sqrt{y - x}$ is

(A) $x = 2(\sqrt{y-x} + \ln|\sqrt{y-x} - 1|) + C$

(B) $x = 2(\sqrt{y-x} - \ln|\sqrt{y-x} - 1|) + C$

(C) $y = 2(\sqrt{y-x} + \ln|\sqrt{y-x} - 1|) + C$

(D) $y = 2(\sqrt{y-x} - \ln|\sqrt{y-x} - 1|) + C$

Q.21

The degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^{2/3} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} = 0$ is

Q.22

If y_1, y_2 are two solutions of the differential eq. $\frac{dy}{dx} + P(x) \cdot y = Q(x)$. Find the sum of constants α,β will the linear combination $\alpha y_1 + \beta y_2$ also be a solution of differential equation.

Q.23

The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi\left(\frac{y}{x}\right)}{\phi'\left(\frac{y}{x}\right)}$ is equal to $\phi\left(\frac{y}{x}\right) = kx^n$ then sum of all possible values of n is

Q.24

The solution of the differential equation $\left(x^2 + 1\right)\frac{dy}{dx} + y^2 + 1 = 0$, is [if $y(0) = 1$]$y = f(x)$ then $f(1) =$

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Q.25

The solution of the equation $\frac{dy}{dx}=\frac{3x-4y-2}{3x-4y-3}$ is $x-y+C=\ln(ax-by+1)$ where a and b are coprime, then $a^2+b^2=$

Q.26

If $y=f(x)$ is the solution of the differential equation $\left(x^2+y^2\right)dy=xydx$ such that $f(1)=1, f\left(x_0\right)=e$, then the greatest integer less than or equal to x_0 is equal to (take $x_0>0$)

Q.27

Suppose $y=y(x)$ satisfies the differential equation $ydx+y^2dy=xdy$. If $y(x)>0\forall x\in\mathbf{R}$ and $y(1)=1$ then $y(-3)$ equals

Q.28

The degree of the differential equation of all curves having normal of constant length C is

Q.29

An inverted conical tank of $2\ \mathrm{m}$ radius and 4 meter height to initially full of water, has an outlet at bottom. The outlet is opened at some instant. The rate of flow through the outlet at any time t is $6h^{3/2}$, where h is height of water level above the outlet at time t. Find the sum of α and β if $t=\frac{\alpha\pi}{\beta}$ (where α and β are in least(+) integer forms).

Q.30

The general solution of the differential equation $\left(3x+2y^2\right)ydx+2x\left(2x+3y^2\right)dy=0$ is $x^ay^b+x^cy^d=C$ then $a+b+c+d=$

Answers & Solutions

Q.1 Answer:

$e^y(x+1)=e^x+c$

Solution:

$(x+1)\frac{dy}{dx}+1=e^{x-y}\Rightarrow(x+1)e^y\frac{dy}{dx}+e^y=e^x$
$\Rightarrow\quad\frac{d}{dx}\left[(x+1)e^y\right]=e^x\Rightarrow(x+1)e^y=e^x+c$

Q.2 Answer:

order 1 and degree 3

Solution:

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We have, $y^2 = 2c(x + \sqrt{c})$ **...(i)**

$\Rightarrow \quad 2yy_1 = 2c \Rightarrow yy_1 = c$ **...(ii)**

Eliminating from (i) and (ii), we get

$$y^2 = 2yy_1\left(x + \sqrt{yy_1}\right)$$

$$\Rightarrow \quad y - xy_1 = \sqrt{yy_1}^{3/2}$$

$$\Rightarrow \quad \left(y - xy_1\right)^2 = yy_1^3$$

Clearly, it is a differential equation of order one and degree 3**.**

Q.3 Answer:

$\frac{d^2x}{dy^2} = 0$

Solution:

The general equation of all non-horizontal lines in xy**-plane is** $ax + by = 1$**, where** $a \neq 0$

Now, $ax + by = 1$

$\Rightarrow a\frac{dx}{dy} + b = 0$ **[Diff.w.r. to** y**]**

$\Rightarrow a\frac{d^2x}{dy^2} = 0$ **[Diff. w.r. to** y**]**

$\Rightarrow \quad \frac{d^2x}{dy^2} = 0$

Hence, the required differential equation is $\frac{d^2x}{dy^2} = 0$

Q.4 Answer:

$xy = \sin x + c\cos x$

Solution:

The given equation can be written as

$\frac{dy}{dx} + \frac{x\sin x + \cos x}{x\cos x}y = \frac{1}{x\cos x}$

which is L.D. equation in y**.**

$$\text{I. F.} \quad = e^{\int \frac{x\sin x + \cos x}{x\cos x}dx} = e^{\int(\tan x + 1/x)dx} = e^{\log\sec x + \log x} = e^{\log(x\sec x)} = x\sec x$$

$\therefore$ **Multiplying the given equation by I.F.** $= x\sec x$**, and integrating, we get**

$$y \cdot x\sec x = c + \int \frac{x\sec x}{x\cos x}dx = c + \int \sec^2 x dx$$

Multiplying by $\cos x$**, this result can also be written as**

$xy = \sin x + c\cos x$ **which is given in (a).**

Hence the correct answers are (a).

Q.5 Answer:

1,2

Solution:

The equation of any tangent to $x^2 = 4y$ **is** $x = my + \frac{1}{m}$**; where** m **is an arbitrary constant.**

Differentiating this w.r. to x**, we get** $1 = m\frac{dy}{dx} \Rightarrow m = \frac{1}{dy/dx}$

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Putting the value of m in $x = my + \frac{1}{m}$, we get $x = \frac{y}{dy/dx} + \frac{dy}{dx} \Rightarrow \left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$ which is a differential equation of order 1 and degree 2.

Q.6 Answer:

$x = y^3 + Ay$

Solution:

Given differential equation is $\left(x + 2y^3\right)\frac{dy}{dx} - y = 0$ or $x + 2y^3 = y\frac{dx}{dy}$

or $\frac{dx}{dy} + \left(-\frac{1}{y}\right)x = 2y^2$, which is linear in x.

$\text{I. F. } e^{\int(-1/y)dy} = e^{-\log y} = \frac{1}{y}$

Hence, the solution is given by

$x\left(\frac{1}{y}\right) = \int 2y^2\left(\frac{1}{y}\right)dy + A = x = y^3 + Ay$

Q.7 Answer:

$y = \pm\sqrt{(\ln x)^2 + c}$

Solution:

$x = e^{\frac{yxdy}{dx}} \Rightarrow \ln x = xy\frac{dy}{dx} \Rightarrow \int ydy = \int \frac{\ln x}{x}dx$

$\frac{y^2}{2} = \frac{(\ln x)^2}{2} + k \Rightarrow y = \pm\sqrt{(\ln x)^2 + c}$

Q.8 Answer:

$\frac{d^3y}{dx^3} = 0$

Solution:

The equation of a member of the family of parabolas having axis parallel to y-axis is

$y = Ax^2 + Bx + C$ **...(i)**

where A, B, C are arbitrary constant.

Differentiating (i) w.r.t. x, we get

$\frac{dy}{dx} = 2Ax + B$ **...(ii)**

which on differentiating w.r.t. x gives

$\frac{d^2y}{dx^2} = 2A \ldots (\text{iii})$

$\text{Differentitating (iii) w.r.t. } x, \text{ we get } \frac{d^3y}{dx^3} = 0$

Q.9 Answer:

none of these

Solution:

Since the equation is not a polynomial in all the differential coefficients so the degree is not defined.

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Q.10 Answer:

$5e^{4x} + 4e^{-5y} = 9$

Solution:

$\frac{dy}{dx} = e^{4x+5y} = e^{4x} \cdot e^{5y}$

$\Rightarrow e^{-5y}dy = e^{4x}dx$

Integrating, we have

$\frac{e^{-5y}}{-5} = \frac{e^{4x}}{4} + C$

$\Rightarrow \quad C = \frac{e^{4x}}{4} + \frac{e^{-5y}}{5}$

Putting $x = 0$ **and** $y(0) = 0, C = \frac{1}{4} + \frac{1}{5} = \frac{9}{20}$

Thus $9 = 5e^{4x} + 4e^{-5y}$

Q.11 Answer:

None of these

Solution:

The given differential equation can be written as

$$x\frac{dy}{dx} + 2y = x(\sin x + \log x)$$

$$\Rightarrow \frac{dy}{dx} + \frac{2}{x}, y = \sin x + \log x$$

which is linear in y **i.e. of the type** $\frac{dy}{dx} + \mathrm{P}y = \mathrm{Q}$

Hence $\mathrm{P} = \frac{2}{x}$

$\therefore \quad \int \mathrm{P}dx = 2\log x = \log x^2$

$\therefore \quad e^{\int \mathrm{P}dx} = e^{\log x^2} = x^2$

$\therefore$ **Sol. is** $y \cdot x^2 = \int x^2(\sin x + \log x)dx + c$

$= -x^2\cos x + 2x\sin x + 2\cos x + \frac{x^2}{3}\log x - \frac{x^2}{9} + c$

i.e. $y = -\cos x + \frac{2}{x}\sin x + \frac{2}{x^2}\cos x + \frac{x}{3}\log x - \frac{x}{9} + \frac{c}{x^2}$

Q.12 Answer:

fixed radius 1 and a variable centres along the x-axis

Solution:

$dx = \frac{y}{\sqrt{1-y^2}}dy$

$\Rightarrow \quad c + x = -\frac{1}{2}\int \frac{2y}{\sqrt{1-y^2}}dy = -\sqrt{1-y^2}$

$\Rightarrow \quad (x+c)^2 + y^2 = 1$

which represents a family of circles of fixed radius 1 and variable centre on the x**-axis.**

Q.13 Answer:

$2\log 18$

Solution:

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The given differential equation in a linear equation with I.F. $= e^{-\int(0.5)dt} = e^{-t/2}$.

Multiplying with I.F. we have

$\frac{d}{dt}\left(e^{-t/2}p\right) = -450e^{-t/2} \Rightarrow e^{-t/2}p(t) = 900e^{-t/2} + C$

When $t = 0, p = 850,$

$850 = 900 + C \Rightarrow C = -50$

Thus $p(t) = 900 - 50e^{t/2}$

When $p(t) = 0$**, we get** $900 - 50e^{t/2} = 0 \Rightarrow \quad e^{t/2} = 18 \quad \Rightarrow \quad t/2 = \log 18$ **i.e.** $t = 2\log 18$

Q.14 Answer:

$x + \cot\left(\frac{x-y}{2}\right) = c$

Solution:

Putting $u = x - y$**, we have** $\frac{\mathrm{d}u}{\mathrm{d}x} = 1 - \frac{\mathrm{d}y}{\mathrm{d}x}$**. The given equation can be written as** $1 - \frac{\mathrm{d}u}{\mathrm{d}x} = \cos u$

$\Rightarrow \quad \int \frac{\mathrm{d}u}{1-\cos u} = \mathrm{d}x$

$\Rightarrow \quad \frac{1}{2}\int \mathrm{cosec}^2\,\frac{u}{2}\,\mathrm{d}u = \int \mathrm{d}x +$ **Const**

$\Rightarrow \quad x + \cot\frac{u}{2} =$ **Const, i.e.** $x + \cot\frac{x-y}{2} = c.$

Q.15 Answer:

$\sin\frac{y}{x} = Cx$

Solution:

The given equation can be written as $y' = \tan\frac{y}{x} + \frac{y}{x}$

This is a homogeneous equation of degree 1 so put $y = Vx.$

$$V + x\frac{dV}{dx} = \tan V + V$$

$$x\frac{dV}{dx} = \tan V \Rightarrow \cot V\,dV = \frac{dx}{x}$$

$\Rightarrow \quad \log\sin V = \log x +$ **Const** $\Rightarrow \sin y/x = Cx$

Q.16 Answer:

$xy + 4 = 0$

Solution:

$x\frac{\mathrm{d}y}{\mathrm{d}x} = -y \Rightarrow \frac{\mathrm{d}y}{y} + \frac{\mathrm{d}x}{x} = 0$

$\Rightarrow \quad \log y + \log x =$ **Const**

$\Rightarrow \quad xy =$ **Const**

Since $y(2) = -2$ **so Const** $= 2(-2) = -4$

Thus $xy + 4 = 0$ **is the required solution**

Q.17 Answer:

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$y = Ae^{kx}$

Solution:

According to the given condition $y\frac{dy}{dx} = ky^2 \Rightarrow \quad \frac{dy}{y} = k\,dx$ **(variables separable equation)** $\Rightarrow$ $\log|y| = kx + C \Rightarrow |y| = Be^{kx} \Rightarrow y = Ae^{kx}$ **where** $A = \pm B$ **and** k **is the constant of proportionality.**

Q.18 Answer:

$-\cos y = 2\sin x \sin^2 y$

Solution:

$$\frac{dy}{dx} = \cos(x-y) - \cos(x+y) = 2\sin x \sin y$$
$$\Rightarrow \quad \frac{dy}{\sin y} = 2\sin x\,dx$$
$$\Rightarrow \quad -\operatorname{cosec} y \cot y = 2\cos x + C$$

Since $y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$ so $C = 0$. Thus

$$-\operatorname{cosec} y \cot y = 2\cos x$$
$$\Rightarrow \quad -\cos y = 2\cos x \sin^2 y$$

Q.19 Answer:

$\left(x^2+y^2\right)^2 = C\left(y^2+2x^2\right)$

Solution:

$(2a-x)2y\frac{dy}{dx} - y^2 = 3x^2$ (Differentiating given equation)

$2(2a-x)y\frac{dy}{dx} = 3x^2 + y^2$

Eliminating a **between the last equation and given equation i.e.** $(2a-x)y^2 = x^2$**, we have**

$y\left(3x^2+y^2\right)\frac{dx}{dy} = 2x^3$

Replacing $\frac{dy}{dx}$ **by** $-\frac{dx}{dy}$ **we have** $2x^3 + y\left(3x^2+y^2\right)\frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{-2x^3}{(3x^2+y^2)y}$**, which is a homogeneous equation.**

Q.20 Answer:

$x = 2(\sqrt{y-x} + \ln|\sqrt{y-x}-1|) + C$

Solution:

$\frac{dy}{dx} = \sqrt{y-x} \quad \Rightarrow \quad \frac{dy-dx}{dx} = (\sqrt{y-x}-1) \quad \Rightarrow \quad \frac{d(y-x)}{(\sqrt{y-x}-1)} = dx$

Let $t = \sqrt{y-x} - 1 \quad \Rightarrow \quad dt = \frac{d(y-x)}{2\sqrt{y-x}}$

$$\Rightarrow \quad d(y-x) = 2(t+1)dt$$
$$\Rightarrow \quad dx = \frac{2(t+1)dt}{t} = 2\left(1+\frac{1}{t}\right)dt$$
$$\Rightarrow \quad x = 2(t + \ln|t|) + c'$$
$$= 2(\sqrt{y-x} - 1 + \ln|\sqrt{y-x}-1|) + c'$$
$$\Rightarrow \quad x = 2(\sqrt{y-x} + \ln|\sqrt{y-x}-1|) + c'$$

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Q.21 Answer:

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Solution:

$$\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} = 0$$

$$\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} = -4 + 3\frac{d^2y}{dx^2} - 5\frac{dy}{dx}$$

$$\left(\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}}\right)^3 = \left(-4 + 3\frac{d^2y}{dx^2} - 5\frac{dy}{dx}\right)^3$$

$$\left(\left(\frac{d^3y}{dx^3}\right)\right)^2 = \left(-4 + 3\frac{d^2y}{dx^2} - 5\frac{dy}{dx}\right)^3$$

The power of $\frac{d^3y}{dx^3}$ is 2 .

Hence, degree of the differential equation is 2 .

Q.22 Answer:

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Solution:

As y_1, y_2 are two solution of D.E.

$\frac{dy}{dx} + P(x)y = Q(x)$

$\frac{dy_1}{ax} + P(x)y_1 = Q(x)$ **...(1)**

$\frac{dy_2}{dx} + P(x)y_2 = Q(x)$ **...(2)**

Now $y = \alpha y_1 + \beta y_2$ will be a solution if

$\frac{d}{dx}(\alpha y_1 + \beta y_2) + P(x)(\alpha y_1 + \beta y_2) = Qx$

$\alpha\left(\frac{dy_1}{dx} + P(x)y_1\right) + \beta\left(\frac{dy_2}{dx} + P(x)\cdot y_2\right) = Q(x)$

$\alpha\{Q(x) + \beta(Q(x)) = Q(x)$

$\alpha + \beta = 1$

Q.23 Answer:

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Solution:

$\frac{dy}{dx} = \frac{y}{x} + \frac{\phi\left(\frac{y}{x}\right)}{\phi'\left(\frac{y}{x}\right)}$. **Put** $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore$ The given differential equation becomes

$v + x\frac{dv}{dx} = v + \frac{\phi(v)}{\phi'(v)} \Rightarrow \frac{\phi'(v)}{\phi(v)}dv = \frac{dx}{x} \Rightarrow \phi\left(\frac{y}{x}\right) = kx$. **(only possible solution i.e.** $n = 1$**)**

Q.24 Answer:

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Solution:

Give $\left(x^2+1\right)\frac{dy}{dx}+\left(y^2+1\right)=0 \quad \int \frac{dy}{y^2+1}=-\int \frac{dx}{x^2+1}$

$\Rightarrow \quad \tan^{-1} y=-\tan^{-1} x+\tan^{-1} c \Rightarrow \tan^{-1} x+\tan^{-1} y=\tan^{-1} c$

$\Rightarrow \tan^{-1} \frac{x+y}{1-xy}=\tan^{-1} c \quad \Rightarrow \frac{x+y}{1-xy}=c$

$\Rightarrow$ Given $y=1$ at $x=0 \Rightarrow \quad \frac{0+1}{1-0}=c \quad \Rightarrow c=1 \Rightarrow \quad \frac{x+y}{1-xy}=1 \quad y=\left(\frac{1-x}{1+x}\right)$

Q.25 Answer:

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Solution:

Put $3x-4y=X \Rightarrow 3-4\frac{dy}{dx}=\frac{dX}{dx} \Rightarrow \frac{dy}{dx}=\frac{1}{4}\left(3-\frac{dX}{dx}\right)$

Therefore the given equation is reduced to

$\frac{3}{4}-\frac{1}{4}\frac{dX}{dx}=\frac{X-2}{X-3} \Rightarrow -\frac{1}{4}\frac{dX}{dx}=\frac{4X-8-3(X-3)}{4(X-3)}=\frac{X+1}{4(X-3)}$

$\Rightarrow -\frac{X-3}{X+1}dX=dx \Rightarrow -\left(1-\frac{4}{X+1}\right)dX=dx$

$\Rightarrow \quad -X+4\log(X+1)=x+$ **constant**

$\Rightarrow \quad 4\log(3x-4y+1)=x+3x-4y+$ **constant**

$\Rightarrow \quad \log(3x-4y+1)=x-y+C.$

Q.26 Answer:

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Solution:

$\frac{dy}{dx}=\frac{xy}{x^2+y^2}$; **Put** $y=vx \quad \left(-\left(\frac{1+v^2}{v^3}\right)\right)dv=\frac{dx}{x}$

$\Rightarrow \quad \frac{1}{2v^2}-\log|v|=\log|x|+c \quad , \quad \therefore \quad \frac{x^2}{2y^2}-\log\left|\frac{y}{x}\right|=\log|x|+c$

As $f(1)=1 \Rightarrow \frac{1}{2}-0 \quad =0+c \therefore \frac{x^2}{2y^2}-\log|y|=\frac{1}{2}$

$\therefore$ **As** $f\left(x_0\right)=e \quad \Rightarrow \quad y_0=e \Rightarrow \frac{x_0^2}{2e^2}-1=\frac{1}{2}, \therefore x_0=\pm\sqrt{3}e \approx 4.70$ **which implies that greatest integer less than or equal to** x_0 **is equal to 4 .**

Q.27 Answer:

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Solution:

$\frac{ydx-xdy}{y^2}=-dy$

$\Rightarrow \quad d\left(\frac{x}{y}\right)=-dy \Rightarrow x/y=-y+c$

since $\quad y(1)=1$ **so** $c=2$ **Thus** $x/y=-y+2$

when $\quad x=-3, -\frac{3}{y}=-y+2$

$\Rightarrow \quad -3=-y^2+2y \Rightarrow (y-1)^2=4$

$\Rightarrow$

$y=1\pm 2=3$ **or** -1

As $y(x)>0$ **for all** $x \in \mathbf{R}$ **so** $y(-3)=3.$

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Q.28 Answer:

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Solution:

According to the given condition

$$y\sqrt{1+\left(\frac{\mathrm{d}x}{\mathrm{d}y}\right)^2}=C \Rightarrow y^2+y^2\left(\frac{\mathrm{d}x}{\mathrm{d}y}\right)^2=C^2$$

The degree of this equation is 2.

Q.29 Answer:

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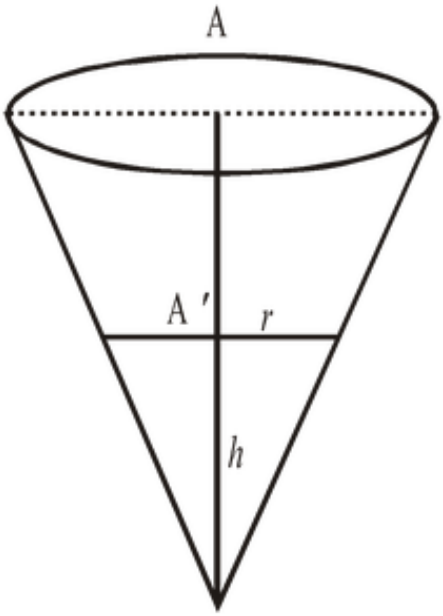
Solution:

Given that $\frac{dv}{dt}=6h^{3/2}$

at time ' t **' the volume of water in the tank is** $v=1/3\pi r^2h$

$\triangle \mathrm{OAB}$ **and** $\mathrm{OA'B'}$ **are similar**

$\frac{\mathrm{OA'}}{\mathrm{A'B'}}=\frac{OA}{AB} \Rightarrow \frac{h}{r}=\frac{4}{2} \Rightarrow r=\frac{h}{2}$



$V=\frac{1}{3}\pi\frac{h^3}{4} \Rightarrow h^3=\frac{12V}{\pi}$

$\Rightarrow \frac{dV}{dt}=\frac{-6\sqrt{12}}{\sqrt{\pi}}\times\sqrt{V}$

$\int_{v_0}^{0}\frac{dV}{\sqrt{V}}=\frac{-6\sqrt{12}}{\sqrt{\pi}}\int_0^t dt \Rightarrow (2\sqrt{V})_{v_0}^{0}=\frac{-6\sqrt{12}}{\pi}=t$

$t=\frac{2\sqrt{V_0}\sqrt{\pi}}{6\sqrt{12}}=\frac{2\sqrt{\pi}}{2\sqrt{12}}\times\sqrt{\frac{16}{3}\pi} \Rightarrow t=\frac{2\pi}{9}$ **unit**

$\alpha+\beta=11$

Q.30 Answer:

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Solution:

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$$
\begin{aligned}
&\left(3x+2y^2\right)ydx+2x\left(2x+3y^2\right)dy=0\\
\Rightarrow\quad &3xydx+2y^3dx+4x^2dy+6xy^2dy=0\\
\Rightarrow\quad &2\left(y^3dx+x.3y^2dy\right)+3xydx+4x^2dy=0\\
\Rightarrow\quad &2d\left(xy^3\right)+3xydx+4x^2dy=0\\
\Rightarrow\quad &2\left(xy^3\right)\cdot d\left(xy^3\right)+3x^2y^4dx+4x^3y^3dy=0\\
\Rightarrow\quad &2\cdot\left(xy^3\right)\cdot d\left(xy^3\right)+d\left(x^3\cdot y^4\right)=0\quad\Rightarrow\quad x^2y^6+x^3y^4=c
\end{aligned}
$$

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